

Abstract Algebra Herstein - Solutions Manual

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Chapter 1

Preliminary Notions

1.1 Set Theory

Problem 1

(a) We expand the definitions $A \subseteq B$ and $B \subseteq C$:

$$x \in A \Rightarrow x \in B$$

$$x \in B \Rightarrow x \in C$$

Suppose $x \in A$. Then by modus ponens, $x \in B$. Again by modus ponens, $x \in C$. Hence, by conditional proof, $x \in A \Rightarrow x \in C$. This is the definition of $A \subseteq C$.

(b) Suppose $x \in A \cup B$. We check two cases. If $x \in A$, then $x \in A$. If $x \in B$, then using $B \subseteq A$ and modus ponens, we get $x \in A$, hence $x \in A$. Thus $A \cup B \subseteq A$.

For the reverse direction, suppose $x \in A$. Then $x \in A \cup B$ by disjunction introduction. Thus $A \subseteq A \cup B$. Hence proven.

(c) Too lazy. Disjunctions are always tedious, as seen previously.

Problem 2

(a) For intersection:

$$x \in A \cap B$$

$$\Leftrightarrow x \in A \text{ and } x \in B$$

$$\Leftrightarrow x \in B \cap A$$

For union: Too lazy. Again, too many disjunctions.

(b) Same idea as (a). Simply apply conjunction elimination twice, then conjunction introduction twice.

Problem 3

Here we gooooo.

Suppose $x \in A \cup (B \cap C)$. If $x \in A$, then $x \in A \cup B$ and $x \in A \cup C$, hence $x \in (A \cup B) \cap (A \cup C)$. Otherwise, $x \in B \cap C$, so $x \in B$ and $x \in C$, thus $x \in B \cup A$ and $x \in C \cup A$, hence $x \in (A \cup B) \cap (A \cup C)$. Hence $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$.

For the reverse direction, suppose $x \in (A \cup B) \cap (A \cup C)$. Then $x \in A \cup B$ and $x \in A \cup C$. If $x \in A$, then $x \in A \cup (B \cap C)$. If $x \in B$, then, in order to satisfy the second statement, either $x \in A$, which we've seen, or $x \in C$. In the latter case, we thus have $x \in B \cap C$, hence $x \in (B \cap C) \cup A$. In all cases, we have $x \in A \cup (B \cap C)$. Hence $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$. Hence proven.

Problem 4

(a)

$$x \in (A \cap B)'$$

$$\Rightarrow x \notin A \cap B$$

Using the fact $\neg(A \wedge B) = \neg A \vee \neg B$:

$$\Rightarrow x \notin A \text{ or } x \notin B$$

$$\Rightarrow x \in A' \text{ or } x \in B'$$

$$\Rightarrow x \in A' \cup B'$$

Too lazy to prove the reverse direction.

(b) Too lazy.

Problem 5

Idk what I'm allowed to do bro.

Problem 6

Include or exclude an element.

Problem 7

At least 39% like both. At most 63% like both.

Problems 8, 9

Too lazy, skipped.

Problem 10

- (a) No. The common ancestor could be different for each pair.
- (b) No. For example, one at far left, one in middle, one at far right.
- (c) Yes.
- (d) Yes.
- (e) No. Equivalence relation must be reflexive.
- (f) Yes.

Problem 11

- (a) The reflexive property guarantees the existence of equivalence classes on non-empty sets, whereas the other properties do not.
- (b) Idk. Maybe $a \in R \Rightarrow a \sim a$.

Problems 12 and 13

Too lazy.

1.2 Mappings**Problem 1**

- (a) Onto, but not one-to-one.
- (b) Both onto and one-to-one. The inverse image is $t\sigma^{-1} = \sqrt{t}$.
- (c) Neither onto nor one-to-one.
- (d) One-to-one, but not onto.

Problem 2

Simply take $f(s \times t) = t \times s$.

Problem 3

Too lazy. Seems obvious.

Problem 4

- (a) Any bijective function has an inverse, which is also a bijection.
- (b) Simply take the composition of the bijection.

Problem 5

For every $s \in S$, we know $\iota(s) = s$, hence:

$$\sigma(\iota(s)) = \sigma(s)$$

Also, since $\sigma(s) \in S$, then identity also applies to $\sigma(s)$, hence:

$$\iota(\sigma(s)) = \sigma(s)$$

Hence,

$$\iota(\sigma(s)) = \sigma(\iota(s)) = \sigma(s)$$

Problem 6

This is Cantor's diagonal argument.

Suppose I have a bijection $f : S \rightarrow S^*$, where each $s_\alpha \in S$ is mapped to a subset $f(s_\alpha) \in S^*$. Let me construct the set $B = \{s \in S \mid s \notin f(s)\}$, which exists. None of the $f(s_\alpha)$ are equal to B . Consider the element s_α . If $s_\alpha \in f(s_\alpha)$, then by definition of B , we have $s_\alpha \notin B$, and if $s_\alpha \notin f(s_\alpha)$, then $s_\alpha \in B$. Hence, $f(s_\alpha)$ and B always differ by whether they contain s_α . Hence, B is a subset of S which is not equal to all $f(s_\alpha)$. Hence, f is not surjective, hence not a bijection. Since for every f , we can find such a B , we conclude that there exists no bijection between S and S^* .

Problem 7

There are $n!$ ways to permute n objects.

Problem 8

(a) If not injective, then two distinct elements will map to one element. Then there remains $n - 2$ elements that must, by surjectivity, map to $n - 1$ elements, which is impossible.

(b) If not surjective, then one element is never mapped to. So we have n elements mapping to $n - 1$ elements, which is impossible. Both (a) and (b) probably require some form of induction, but I'm too lazy.

(c) For (a), consider the map $f : \mathbb{R} \rightarrow \mathbb{R}$ composed of $[0, 1) \rightarrow \mathbb{R}$, and $[1, 2) \rightarrow \mathbb{R}$, and $[2, 3) \rightarrow \mathbb{R}$, and so on (and also in the negative direction). In real analysis, we learned that there is bijection between every half-open interval and $\mathbb{R}$. For (b), simply map $\mathbb{R}$ to $(0, 1)$, also borrowing a bijection from real analysis.

Problem 9

(a) Using the real numbers again, let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ and $\tau : (0, 1) \rightarrow \mathbb{R}$, but the range of σ is $(0, 1)$.

(b)

$$\sigma : X \rightarrow Y$$

$$\tau : Y \rightarrow Z$$

Consider $Y - \sigma(X)$. Notice that τ needn't be injective on $Y - \sigma(X)$. Because the composition only cares about the behavior of $X \rightarrow \sigma(X) \rightarrow \tau(\sigma(X))$.

Problem 10

Classic real analysis exercise.

Problem 11

- (a) $a\sigma_A = a\sigma \in T$
- (b) Suppose $a, b \in A$ and $a \neq b$. We know $a, b \in S$, hence $a\sigma \neq b\sigma$ which is equivalent to $a\sigma_A \neq b\sigma_B$.
- (c) Obviously, σ_A could be the injective portion of σ .

Problem 12

Given $a \in A$, then $a\sigma \in A$. Hence, we can apply the restricted version σ_A on $a\sigma$. Thus

$$(a\sigma)\sigma = (a\sigma)\sigma_A$$

Since $a \in A$, then $a\sigma = a\sigma_A$, which we use to change the right side of the equation:

$$(a\sigma)\sigma = (a\sigma_A)\sigma_A$$

This is equivalent to

$$a(\sigma \circ \sigma) = a(\sigma_A \circ \sigma_A)$$

Since $a \in A$, we can restrict the left side to only

$$a(\sigma \circ \sigma)_A = a(\sigma_A \circ \sigma_A)$$

Problem 13

- (a) Let $\sigma : n \mapsto 2n$.
- (b) Let $\sigma : \mathbb{R} \rightarrow (0, 1)$, borrowing a function from real analysis.
- (c) A is infinite, so it has a map $f : A \rightarrow a$, where $a \in A$. Then, we use the identity function on $S \setminus A$. Combining the two functions, we effectively get a bijection $g : S \rightarrow s$ where $s \subset S$:

$$g(x) = \begin{cases} f(x) & x \in A \\ i(x) & x \in S \setminus A \end{cases}$$

Problem 14

We already know there exists a bijection $f : \mathbb{Z} \leftrightarrow \mathbb{Z}^2$. If there exists a bijection $g : S \leftrightarrow \mathbb{Z}$, consider the function $h : \mathbb{Z}^2 \rightarrow S^2$ defined by $h(x \times y) = g^{-1}(x) \times g^{-1}(y)$. Using the bijectivity of g , it can be shown that h is bijective. The bijection we were looking for is the composition $g \circ f \circ h : S \leftrightarrow S^2$.

Problem 15

If $f : T \rightarrow S$ and $g : U \rightarrow T$ are surjective, then the composition $g \circ f$ is surjective. I don't know how to prove that there exists no surjection in the other direction.

Problem 16

Too lazy.

1.3 The Integers**Problem 1**

$$a \mid b \Rightarrow b = k_1 a$$

$$b \mid a \Rightarrow a = k_2 b$$

Where k_1, k_2 are integers. But we know $k_2 = \frac{1}{k_1}$, and in order for k_2 to remain an integer, k_1 must be equal to 1 or -1 (this can be proved by induction). Hence, $b = (\pm 1)a = \pm a$.

Problem 2

$$b \mid g, \quad b \mid h \Rightarrow b \mid mg, \quad b \mid nh \Rightarrow b \mid mg + nh$$

Problem 3

Let $d = \frac{ab}{(a,b)} = \frac{a}{(a,b)}b$. Since $(a,b) \mid a$, then $\frac{a}{(a,b)}$ is an integer, hence $b \mid d$. The same procedure can be done for $a \mid d$. Hence, d satisfies the first property.

For the second property, first note that if $a = m(a,b)$ and $b = n(a,b)$, then $(m,n) = 1$. Suppose for contradiction that $k \mid m$ and $k \mid n$ for some $k > 1$. Then

$$k(a,b) \mid m(a,b) = a$$

and

$$k(a,b) \mid n(a,b) = b$$

hence $k(a,b) \mid a, b$ but $k(a,b) \nmid (a,b)$, contradicting the fact that (a,b) is the GCD. Hence $(m,n) = 1$.

SKIP.

Problem 4

$$am = bn$$

Since $(a,b) = 1$, and $a \mid bn$, then $a \mid n$. Hence $ab \mid nb = x$.

Problem 5

(a) Obviously, $p_1^{\delta_1} \dots p_k^{\delta_k}$ divides both a and b . Now, suppose $x \mid a, b$. We can develop its prime factorization. So $x = p_1^{\omega_1} \dots p_k^{\omega_k} \mid a, b$. Which means $p_i^{\omega_i} \mid a, b$, for each i . But $p_i^{\omega_i}$ is coprime with every other prime of a, b , so it must divide the non-coprime part, which only consists of the prime equal to him, hence $p_i^{\omega_i} \mid p_i^{\alpha_i}$ and $p_i^{\omega_i} \mid p_i^{\beta_i}$. Hence, it must divide the minimum power, so $\omega_i \mid \min(\alpha_i, \beta_i) = p_i^{\delta_i}$. Multiplying every $p_i^{\omega_i}$ together, we get $p_1^{\omega_1} \dots p_k^{\omega_k} \mid p_1^{\delta_1} \dots p_k^{\delta_k}$, which gives us our desired result.

(b) We use the formula proven in Problem 3.

$$[a, b] = \frac{ab}{(a, b)}$$

Let's concentrate on a single prime at a time. For prime i , we have

$$\begin{aligned} p_i^{\gamma_i} &= \frac{p_i^{\alpha_i} p_i^{\beta_i}}{p_i^{\min(\alpha_i, \beta_i)}} \\ &= p_i^{\alpha_i + \beta_i - \min(\alpha_i, \beta_i)} \\ &= p_i^{\max(\alpha_i, \beta_i)} \end{aligned}$$

Problem 6

Proof by induction. Base case. r_{n-1} is divisible by r_n .

$$r_{n-1} = q_n r_n + r_{n+1} = q_n r_n + 0 = q_n r_n$$

And obviously, r_n is divisible by r_n .

Now, suppose r_{k-1} and r_k are divisible by r_k , then

$$r_{k-2} = q_{k-1} r_{k-1} + r_k$$

indicates that r_{k-2} is obviously divisible by r_k . Here, we thus proved:

$$r_k \mid r_k \text{ and } r_k \mid r_{k-1} \quad \Rightarrow \quad r_k \mid r_{k-1} \text{ and } r_k \mid r_{k-2}$$

which completes our induction step. (a and b are precisely $r_{(-1)}$ and r_0 .)

Problem 7

Nope.

Problem 8

First, if it's not divisible by any prime below n , since every composite below n can be written as multiple of some smaller prime, then the prime not being a divisor means that the composite is not a divisor as well. Hence, checking for prime divisibility already guarantees that the composites are not divisors.

Second, if p divides n , then we can write $pq = n$ for some q . If $p > \sqrt{n}$, then $\frac{n}{p} < \frac{n}{\sqrt{x}}$ which is equivalent to $q < \sqrt{n}$. Hence, for every integer $p > \sqrt{n}$, it suffices to check for some other $q < \sqrt{n}$.

Chapter 2

Group Theory

2.1 Some Preliminary Lemmas

Problem 1

- (a) No. Not associative
- (b) No. Missing inverse.
- (c) Yes.
- (d) Yes.

Problem 2

Just repeatedly swap.

Problem 3

For all a, b , we have

$$(a \cdot b)^2 = a^2 \cdot b^2$$

Developing both sides, we have

$$a \cdot b \cdot a \cdot b = a \cdot a \cdot b \cdot b$$

Take a inverse on left, and b inverse on right.

$$\Rightarrow b \cdot a = a \cdot b$$

Problem 4

We are given (1)

$$(ab)^i = a^i b^i$$

and (2)

$$(ab)^{i+1} = a^{i+1} b^{i+1}$$

$$\Rightarrow (ba)^i = a^i b^i$$

and (3)

$$\begin{aligned} (ab)^{i+2} &= a^{i+2} b^{i+2} \\ \Rightarrow (ba)^{i+1} &= a^{i+1} b^{i+1} \end{aligned}$$

We substitute (2) into (3), then regroup, giving the result (4)

$$\begin{aligned} \Rightarrow (ba)^{i+1} &= a(ba)^i b \\ \Rightarrow ba(ba)^i &= ab(ab)^i \end{aligned}$$

Notice that we can equate $a^i b^i$ in (1) and (2), giving the result (5)

$$(ab)^i = a^i b^i = (ba)^i$$

Now, we substitute (5) into (4), which becomes

$$\Rightarrow ba(ba)^i = ab(ba)^i$$

Taking the inverse on each side gives the desired result.

Problem 5

Too dumb for this.

Problem 6

Let $x = 1 \leftrightarrow 2$. Let $y = 2 \leftrightarrow 3$. Then $(xy)^2$ will cause $123 \rightarrow 312$, while $x^2 y^2$ would be identity $123 \rightarrow 123$.

Problem 7

The identity operation, plus the 3 swaps, satisfy $x^2 = e$. Then, the identity, plus the 2 cycles satisfy $x^3 = e$.

Problem 8

Suppose for contradiction that $a^N \neq e$ for all $N \in \mathbb{N}$. We show that this implies $a^X \neq a^Y$ for all pairs $X \neq Y$. Suppose for (nested) contradiction that $a^X = a^Y$ for some pair $X > Y > 1$. Then taking the inverse on both sides yields $a^{X-Y} = e$, contradicting the fact that $a^N \neq e$ for all N . Hence, $a^X \neq a^Y$, i.e. the powers of a are pairwise distinct. However, we know our group is finite, but a^N creates infinite distinct elements. Hence, it must be false, so $a^N = e$ for some N .

Problem 9

(a)

$$G = \{e, x, y\}$$

Since identity e is unique, then $xy \neq x$. Because, otherwise, we would have $y = e$, which is false. Similarly, $xy \neq y$. So, using the fact that groups are closed, xy must have some result in G , the only one left being $xy = e$. Property of inverse implies that $yx = e$ as well. Hence x commutes with y . Finally, we know the identity commutes with every elements. Hence proven.

(b)

$$G = \{e, x, y, z\}$$

Suppose $xy = yx = e$, then xz cannot equal x nor z , because neither is equal to the unique e . It also cannot equal e , because x, y are already inverse pairs, and inverses are unique, so z cannot be another inverse. Hence xz can only equal y , by the closure property. This procedure is repeated for zx, yz, zy , which must equal y, x, x respectively. Hence, $xy = yx = e$ and $xz = zx = y$ and $yz = zy = x$, where e already commutes with everyone.

Now, suppose $x^2 = e$ and $y^2 = e$. Since inverses are unique, we have $z^2 = e$. And thus $xy = yx = z$ and $xz = zx = y$ and $yz = zy = x$ (same logic as before, none can be identity, and none is inverse pair). And e already commutes with everyone.

All cases have been gone over. Hence proven.

(c) We know every group of prime order must be cyclic, therefore abelian. (Using later results in book, because I'm too dumb)

Problem 10

For every pair x, y , we have

$$(xy)^2 = e = ee = x^2y^2$$

$$\Rightarrow xyxy = xxyy$$

Taking inverse on both extremities of each side yields

$$yx = xy$$

Problem 11

Inverses come in pairs. But e is paired with itself. So if the group order is even, then there is an odd number of non- e elements. Hence, another element must be paired with itself.

Problem 12

Inverse is commutative:

$$ay = e$$

$$yay = ye$$

$$yay = y$$

$$yayy^{-1} = yy^{-1}$$

$$ya = e$$

We use this result to prove identity is commutative:

$$ae = a$$

$$aya = a$$

$$ea = a$$

Problem 13

I'm too dumb.

Problem 14

Since the group is finite, for every a , there exists m such that $a^m = a$. Let $e_a = a^{m-1}$ be an identity of a . It's clear by associativity that it's both left and right identity of a .

Let $a^{-1} = a^{m-2}$ be an inverse of a . It's also clear by associativity that it's both left and right inverse of a .

Suppose $ab = e_a$. Recall that $a \cdot a^{m-2} = a^{m-1} = e_a$. Hence,

$$ab = e_a = a \cdot a^{m-2}$$

$$\Rightarrow b = a^{m-2}$$

by left cancellation law. Thus, each element's inverse is unique. The same procedure also shows that each element's identity is unique.

Suppose $ae_a = a$. Then

$$ae_a c = ac$$

$$\Rightarrow e_a c = c$$

by left cancellation law. Hence, every element shares an identity pairwise. Since every element also only has a single identity, then all identities are equal.

Problem 15

(a) It's already finite, closed, and associative. Suppose for contradiction that $ab \equiv ac \pmod{p}$, but $b \not\equiv c \pmod{p}$, where $a, b, c \in \{1, \dots, p-1\}$. Then $ab - ac = a(b - c) \equiv 0 \pmod{p}$. Hence $p \mid a(b - c) = am$ where clearly $m < p$. Now, since p is prime, it must divide one of the factors a or m . But $a < p$ and $m < p$, thus we have a contradiction. Hence, it must be that $b \equiv c \pmod{p}$. Hence, by Problem 14, this is a group.

(b) Skip

Problem 16

I'm too dumb.

Problem 17

Consider the naturals $\mathbb{N} = \{1, 2, \dots\}$ under addition. It's associative and closed, and the cancellation laws work. However, the latter still don't allow us to dip lower than 1. Hence, we don't have access to the identity element.

Problem 18

S_3 are the reflections/rotations of an equilateral triangle. We can simply look at higher n -gons (squares, regular pentagons, etc). Each has n rotations (including identity), and 2 reflections (including identity), which is a total of $n \cdot 2 = 2n$ group elements. The triangle being the smallest regular polygon, it makes sense that $n > 2$.

Problem 19

Nah.

Problem 20

It is closed. Suppose $\det(A), \det(B) \in \mathbb{Q}$. Then,

$$\det(AB) = \det(A)\det(B) \in \mathbb{Q}$$

It is associative, because matrix multiplication is inherently associative. There exists the identity matrix which has determinant 1, and clearly $1 \in \mathbb{Q}$. And since determinant is nonzero, then there exists an inverse for each matrix.

Problem 21

It is closed:

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \begin{pmatrix} e & f \\ 0 & h \end{pmatrix} = \begin{pmatrix} ae & af + bh \\ 0 & dh \end{pmatrix}$$

and since $ad, eh \neq 0$, then $adeh = (ae)(dh) \neq 0$. Matrix multiplication is inherently associative. The identity matrix satisfies the conditions. And since ad is the determinant, which is nonzero, then every matrix has an inverse. Finally,

$$\begin{pmatrix} e & f \\ 0 & h \end{pmatrix} \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} ae & be + df \\ 0 & dh \end{pmatrix}$$

so it is not abelian.

Problem 22

Clearly closed and associative. The identity satisfies the conditions also. Next, since a is nonzero, then a^{-1} exists and is also nonzero. Therefore their product is nonzero. And $aa^{-1} = 1$ is precisely their determinant, which is nonzero, hence the inverse exists. Finally, diagonal matrices are commutative, hence the group is abelian.

Problem 23

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

Problem 24

We can simply make an exhaustive list:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

It contains 6 elements. The identity is present. If you check all cases, you will see that it is closed and each element has an inverse. Matrix multiplication is already associative.

Problems 25-26

Too lazy.

2.2 A Counting Principle

Problem 1

Associativity is inherited. If $x, y \in H \cap K$, then $xy \in H$ and $xy \in K$, hence $xy \in H \cap K$. Both contain e . If $x \in H \cap K$, then $x^{-1} \in H$ and $x^{-1} \in K$, hence $x^{-1} \in H \cap K$.

Problem 2

$\langle a \rangle$ is a subgroup. But $\langle a^2 \rangle$ is also a subgroup. They share an element $a^m = x = (a^2)^n$ for some m, n . Hence, $a^{m-2n} = e$, which gives an order of $o(a) = m - 2n$. In the case that $m = 2n$, then try higher prime powers, such as $\langle a^3 \rangle, \langle a^5 \rangle$, and so on. It's impossible for m to contain all prime factors, so eventually, there will be some $\langle a^p \rangle$ for which $p \nmid m$.

Problem 3

Which means the only subgroups are $\{e\}$ and G itself. So if you take any element $a \in G$, then $\langle a \rangle = G$ or $\langle a \rangle = \{e\}$. The latter case simply means $a = e$.

Now, in the first case, suppose for contradiction that $\langle a \rangle = G$ is infinite. Since G is a group, then for every a^m , it has an inverse a^n . Thus, for every $(a^2)^m$, it has an inverse $(a^2)^n$, because $(a^2)^m (a^2)^n = (a^m a^n)^2 = e^2 = e$. Hence, $\langle a^2 \rangle$ forms a nontrivial subgroup of G , contradicting the requirement. Therefore, G must be finite.

Now, suppose for contradiction that G isn't of prime order. Then take any divisor of $p \mid o(G)$. Clearly, $\langle a^p \rangle$ is a nontrivial subgroup of G , contradicting the requirement. Hence, G must be of prime order.

Problem 4

(a) Suppose $ah_1a^{-1}, ah_2a^{-1} \in aHa^{-1}$, then

$$(ah_1a^{-1})(ah_2a^{-1}) = ah_1h_2a^{-1} = ah_3a^{-1} \in aHa^{-1}$$

Now, since $e \in H$, then

$$aea^{-1} = aa^{-1} = e \in aHa^{-1}$$

Finally, since H is a subgroup, then h^{-1} exists for every h . So for aha^{-1} , choose $ah^{-1}a^{-1}$ as its inverse. We verify:

$$(aha^{-1})(ah^{-1}a^{-1}) = ah h^{-1} a^{-1} = aea^{-1} = e$$

(b) Given $h_1, h_2 \in H$, if $h_1 \neq h_2$, then $ah_1 \neq ah_2$, and then $ah_1a^{-1} \neq ah_2a^{-1}$. (These follow from left and right cancellation laws.) Therefore, $o(aHa^{-1}) = o(H)$.

Problem 5

We assign $ha \rightarrow ah$ for every $h \in H$. Clearly, this is surjective. Now, if $ah_1 = ah_2$, then by cancellation law $h_1 = h_2$, hence $h_1a = h_2a$, thus the map is also injective.

Problem 6

- (a) We have H and Ha .
 (b) We have H, Ha, Ha^2, Ha^3 and Ha^4 .
 (c) Here, H contains all permutations that leave x_1 unmoved. Let a be the transformation that shifts everything to the right once. Let b be the transformation that shifts everything to the right twice. Then the cosets are H, Ha and Hb . These correspond to fixing x_1 at the first position, then fixing x_1 at the second position, then fixing x_1 at the third position.

Problem 7

Both (a) and (b) are cyclic groups, so they remain the same by their abelian property. For (c), we keep the same transformations a and b . However, now it becomes fixing x_1 at the first position, then fixing x_2 at the first position, then fixing x_3 at the first position.

Problem 8

For (a) and (b), yes, because they are abelian. For (c), no, as we demonstrated.

Problem 9

(Searched it up: It uses the concept of normal groups discussed later.)

The condition can be expressed in another way: if $aH = bH$ then $Ha = Hb$. Let b be the inverse of some element c . The hypothesis should still hold: $aH = c^{-1}H$ then $Ha = Hc^{-1}$.

Now, if $aH = c^{-1}H$, then

$$\begin{aligned} a &\in c^{-1}H \\ \Rightarrow a &= c^{-1}h \text{ for some } h \in H \\ \Rightarrow ca &= h \in H \end{aligned}$$

Similarly, if $Ha = Hc^{-1}$, then

$$\Rightarrow ac \in H$$

Hence, the hypothesis is equivalent to: $ca \in H$ implies $ac \in H$.

Now, suppose $x = ghg^{-1} \in gHg^{-1}$ for some $h \in H$. Then $g^{-1}xg = h \in H$. We use the "abelian"-ish property of the hypothesis to switch their positions:

$$g^{-1}(xg) \in H \Rightarrow (xg)g^{-1} \in H$$

which produces the desired result $x \in H$.

Problem 10

n cosets, each of the form $H_n + x$ where $x \in \{0, 1, \dots, n-1\}$.

Problem 11

$$H_n \cap H_m = H_{\text{lcm}(n,m)}$$

Problem 12

Consider the cosets of $H \cap K$ in G , for example $(H \cap K)g$. The following holds:

$$(H \cap K)g = (Hg \cap Kg)$$

Since H and K have finite index in G , then the total number of intersections of their cosets is finite as well, namely $Ha_i \cap Kb_j$, which include $(Hg \cap Kg)$. The cosets of $H \cap K$ are each equal to one of these finitely many intersections, hence $H \cap K$ has finite index in G .

Problem 13

If $x, y \in N(a)$, then $xya = xay = axy$, hence $xy \in N(a)$. Of course, $e \in N(a)$, since the identity commutes with everything. Finally, if $x \in N(a)$, then

$$x^{-1}a = x^{-1}axx^{-1} = x^{-1}xax^{-1} = ax^{-1}$$

Problem 14

Same as problem 13.

Problem 15

Same as problem 13. And $Z = C(G)$.

Problem 16

(a) If $a, b \in N(H)$, then $abH(ab)^{-1} = abHb^{-1}a^{-1} = aHa^{-1} = H$, hence $ab \in N(H)$. Of course, $eHe^{-1} = eHe = H$. Finally, suppose $a \in N(H)$. We show $a^{-1}Ha \subseteq H$. Given $h \in H$, we have

$$\begin{aligned} a^{-1}ha \\ &= (ah^{-1}a^{-1})^{-1} \\ &= (h')^{-1} \in H \end{aligned}$$

where h' is some element in H . Implicitly, this also shows the reverse inclusion. For any given $h \in H$, we can simply express it as the inverse of some other element h' , and work our way back the equality to show . Hence, proven.

(b) Obviously, if $a \in H$, then by closure, we have $aha^{-1} \in H$, hence $a \in N(H)$.

Problem 17

If $x \in C(H)$, then $xh = hx$ for all $h \in H$, then $xhx^{-1} = h$, hence $x \in N(H)$.

As for the example, consider G the dihedral group, with H being the subgroup of rotations. Here, $N(H) = G$, because if a is a rotation, then the resulting aha^{-1} is still a rotation, and if a is a reflection, then aha^{-1} reflects twice, which preserves the orientation, so it's still a rotation.

However, $C(H) \neq G$, because the order of application of rotations and reflections matters, hence the reflections are not included. The important distinction is that $C(H)$ requires both sides to be equal ($ah = ha$), whilst $N(H)$ only requires inclusion ($ah \in Ha$).

So we can say that $C(H) \subseteq N(H)$.

Problem 18-28

To be completed.

Problem 29

Coprimes are: 1, 3, 5, 7. Since $3^2 = 9 \equiv 1 \pmod{8}$, hence U_8 is not cyclic.

Problem 30

Coprimes are: 1, 2, 4, 5, 7, 8.

$2^2 = 4$; $2^3 = 8$; $2^4 = 16 \equiv 7 \pmod{9}$; $2^5 = 32 \equiv 5 \pmod{9}$; $2^6 = 64 \equiv 1 \pmod{9}$. Hence, U_9 is cyclic with generator 2.

Its other generator is: 5.

Problem 31**2.3 Normal Subgroups and Quotient Groups****Problem 1**

Consider the cosets Hg and Hg^{-1} . By hypothesis, $HgHg^{-1} = Hx$ for some coset Hx . The left side contains e (since $egeg^{-1} = e$), so the right side also contains e . Thus, it must be that $Hx = H$, because the subgroup itself is its only coset that contains e . Hence, $HgHg^{-1} = H$. Now, since $e \in H$, then $egHg^{-1} \subseteq H$, which simplifies to $gHg^{-1} \subseteq H$.

Problem 2

We use the disjoint property of cosets. Suppose $x \notin H$. Both Hx and xH share the same complement H :

$$G - Hx = H = G - xH$$

$$\Rightarrow Hx = xH$$

Problem 3

Suppose we have two elements of NH , namely n_1h_1 and n_2h_2 . We derive:

$$(n_1h_1)(n_2h_2) = n_1(h_1n_2)h_2 = n_1(n_2h_1)h_2$$

which is contained in NH . The second equality used the fact that hN is also a right coset Nh . So closure is satisfied.

Of course, NH contains $ee = e$.

Finally, we know $NH = HN$. If $nh \in NH$, i.e. $h \in H$ and $n \in N$, then by subgroup property, $h^{-1} \in H$ and $n^{-1} \in N$, hence $h^{-1}n^{-1} \in HN = NH$.

Problem 4

Let $N_1, N_2 \subseteq G$, and $N = N_1 \cap N_2$. Then,

$$gNg^{-1} = g(N_1 \cap N_2)g^{-1} \subseteq gN_1g^{-1} = N_1$$

$$gNg^{-1} = g(N_1 \cap N_2)g^{-1} \subseteq gN_2g^{-1} = N_2$$

Hence, $gNg^{-1} \subseteq N_1 \cap N_2 = N$.

Problem 5

Intersection of subgroups is still a group, and since $H \cap N$ is subset of H , it follows that $H \cap N$ is a subgroup of H .

Now, since N is normal in G , then for every $h \in H \subseteq G$, we have

$$hNh \subseteq N$$

which implies

$$\Rightarrow h(H \cap N)h \subseteq N$$

Also, since H is a subgroup, and the left side is the product of elements in H , we have

$$\Rightarrow h(H \cap N)h \subseteq H$$

Hence,

$$h(H \cap N)h \subseteq H \cap N$$

for every $h \in H$, thus $H \cap N$ is normal in H .

Problem 6

If H is a subgroup, then $ghg^{-1} = hgg^{-1} = h \in H$, hence H is normal.